

May, 2026

SE 4458 Software Architecture & Design of Modern Large Scale Systems - Final

Group 1: Create a Hotel Booking System

Create a hotel booking system like Hotels.com. Implement Below are use cases for functional and non-functional requirements.

HOTEL ADMIN SERVICE



Hotel Swiss

Baslangic 

Bitis 

Oda Tipi: 

Oda Adedi

☐ Dolu ☒ Bos

- Admins for hotels can add/update rooms for availability between start and end dates
- Image uploading is not necessary but nice-to-have
- THIS WILL BE AN AUTHENTICATED SERVICE

HOTEL SEARCH SERVICE



Hotels.com Seyahat Ara  Baris

Nereye?

Varis noktası
Bodrum, Muğla, Türkiye

Tarihler
12 May - 16 May

Misafir sayısı
2 misafir, 1 oda

Haritada göster

300 veya daha fazla konaklama yeri

Konaklama yeri sıralamamızın işleyiş şekli

Sıralama ölçütü:

Önerilen

Giriş yaptığınızda binlerce otelde ortalama %15 oranında kazanç sağlarsınız

Giriş yap

Hyde Bodrum - Yetişkin Oteli

Torba

Her şey dâhil Havuz

43.791 TL

4 gece için

Gecelik 10.948 TL

vergiler ve ücretler dâhildir

Puan toplayın

10 Olağanüstü 3 yorum

Ekstra kazanç için hesabınıza giriş yapın

MGallery The Bodrum Hotel Yalıkavak

Yalıkavak

Havuz

37.830 TL

4 gece için

Gecelik 9.458 TL

vergiler ve ücretler dâhildir

Puan toplayın

9,6 Olağanüstü 30 yorum

Staff Agent

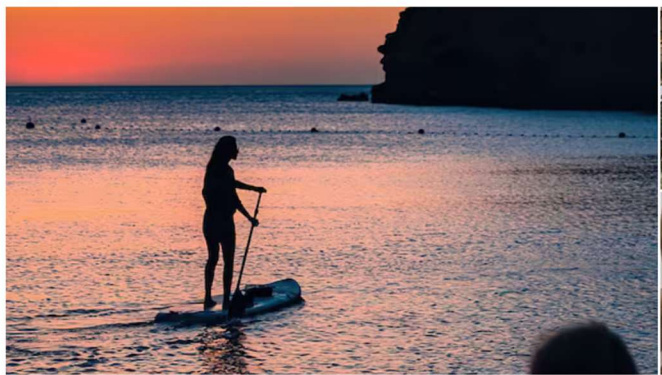
You are chatting with an AI Agent.

Write a message...

Send

- User can search hotels by destination point, dates and number of people. Only rooms that admins specified as vacant will appear in results for given dates
- Client who login to application will see 15% discounted prices.
- 'Haritada goster' is required to show hotels that have been searched

BOOK HOTEL SERVICE



Genel Bakış Otel Özellikleri Odalar Engellilere Yönelik Özellik Politika

Hyde Bodrum - Yetişkin Oteli

8,8 Mükemmel

32 yorumun tümünü göster

Üye Fiyatı: %10 indirim

42.225 TL

~~46.917 TL~~

4 gece için

Gecelik 10.556 TL

vergiler ve ücretler dâhildir

Fiyat bilgileri

Rezervasyon yap

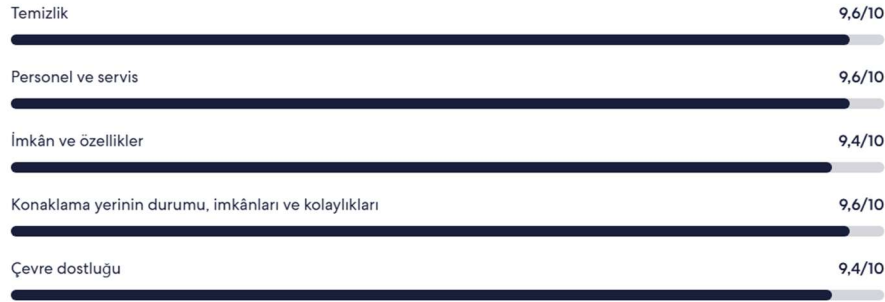
- On hotel detail page, user can book the hotel. Capacity of the hotel needs to be decreased for specified dates
- NO transaction needed for payment

HOTEL COMMENTS SERVICE

- On clicking of comments, a graphs that shows distribution of comments per service will be shown along with comments and ratings

9,6/10 Olağanüstü

163 doğrulanmış yorum ⓘ



8/10 Çok İyi

16 Haziran 2025



Simge, 4 gecelik seyahat

16 Haziran 2025 tarihinde Enver tarafından gönderilen yanıt

Geri bildiriminiz için teşekkür ederiz.

NOTIFICATION SERVICE

- write a nightly scheduled task
 - o to go over all hotel capacities and notify hotel administrators when it is below 20% for the next month.
 - o to pull new hotel reservations from the queue and send them a message about reservation details

AI AGENT SERVICE

Implement AI agent chat window in main application screen that will use APIs you created to perform search and booking use cases

You: I'd like to book a hotel in Rome from July 15 to July 18 for two adults.

AI: Great! Any preferences for hotel rating, price range, or amenities?

You: 4+ stars, breakfast, and a pool would be ideal.

AI: Got it! Here are a few options you might like 🙌

Hotel Roma Plaza

- 📍 City Centre
- ⭐ 4.5 | 💰 \$210/night
- ✅ Free Wi-Fi, Breakfast, Pool

- ○ [Reserve Room]

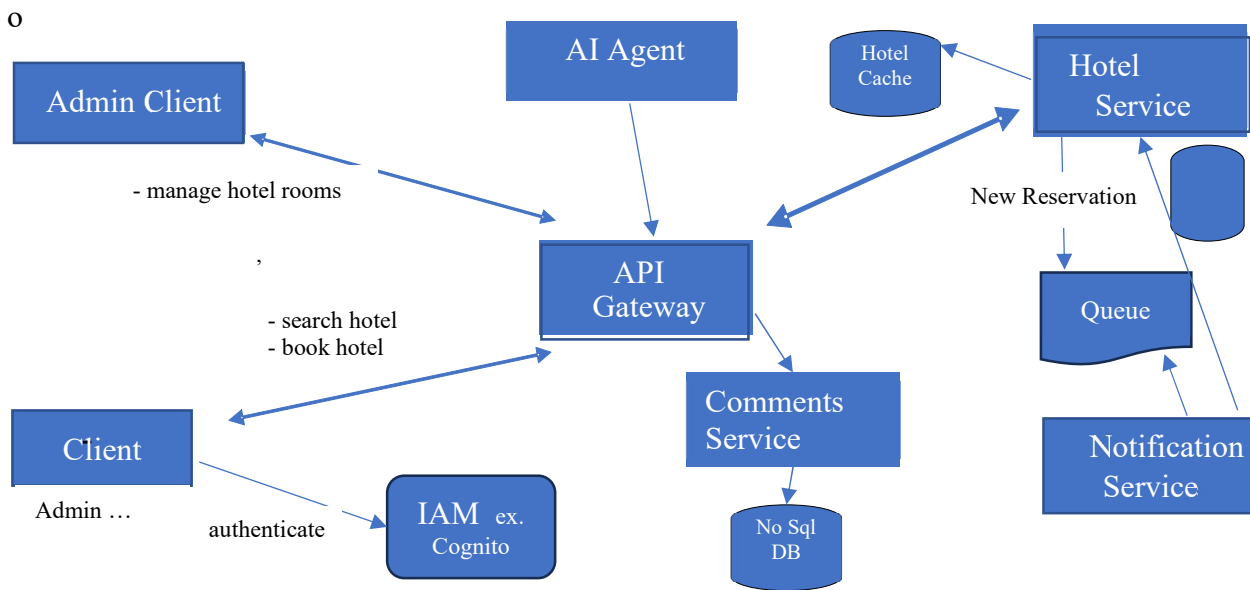
 **Grand Hotel**

-  Monti
-  4.3 |  \$250/night
-  Free Wi-Fi, Breakfast, Pool
- ○ [Reserve Room]

 AI: Would you like to confirm your reservation at Hotel Roma Plaza from July 15–18 for 2 adults?

○ Yes, book it

NON-FUNCTIONAL REQUIREMENTS



- Comments will be stored in a separate No SQL DB
- Hotel details will be stored in a separate distributed cache like Redis

Students

Batuhan Salcan	Ayfernaz Baygin
Mustafa Berkay Duzenli	SÜMEYYE ŞENCAN
Batikan Akdeniz	BEGÜM BAL
Cenk Serbest	EREN KARCI
Toprak Orman	BARIŞ HANSU
Ilayda Gun	Idil Balandi
Demir Demirdogen	Berk Ates
Aycan Kurt	

Group 2 – Create a job search web application

Create a job search site web application like kariyer.net that will have following use cases/screens

ADMIN SERVICE

Authorized Admins and companies can add/update jobs

Yazılım Uzmanı

Alfemo

Kaydet

Izmir(Torbalı) • İş Yerinde

Çalışma Şekli

Tam Zamanlı

Pozisyon Seviyesi

Uzman

Departman

Bilgi Teknolojileri / IT

Yazılım Uzmanı

Alfemo

- Üniversitelerin ilgili bölümlerinden mezun (Bilgisayar Mühendisliği, Yazılım Mühendisliği veya Bilgisayar Programcılığı).
- HTML5 , CSS3, Bootstrap, Javascript, JQuery, Ajax, JSON, XML bilgisine sahip.
- Microsoft .NET platformu, C#, ASP.NET MVC ile uygulama geliştirme konusunda bilgi ve tecrübe sahibi.
- Tercihen MS SQL Server konusunda bilgi sahibi.
- Birimler arası entegrasyon süreçleri hakkında bilgisi olan.

Daha Fazla Gör

Aday Kriterleri

Tecrübe: En az 2 yıl tecrübeli

Eğitim Seviyesi: Üniversite(Mezun), Yüksek Lisans(Öğrenci), Yüksek Lisans(Mezun)

Askerlik Durumu: Yapıldı, Muaf

HOME PAGE

kariyer.net

EN

Giriş Yap / Üye Ol

Kariyer Fırsatlarını Keşfet

65.200 iş ilanı on binlerce şirket

Pozisyonlarda Ara

Yazılım Uzmanı

Satın Alma Müdürü

Muhasebe Uzmanı

Şehir veya ilçe ara

İş Bul

Yazılım Geliştirme Uzmanı

Proje Yöneticisi

İK Uzmanı

Ankara

Izmir

Bursa

Istanbul

Staff Agent

You are chatting with an AI Agent

Write a message...

Send

Son Aramalarım

Izmir – Web Developer – Fulltime

Tum Turkiye – Full Stack Developer

ÖNE ÇIKAN İLANLAR

 Perakende Kariyer Günlüğü Kariyer Günlüğü İstanbul (Tümü) +10 il daha	 Birikim Güvence Danışmanı - Marmaris Agesa Muğla	 Kurumsal Satış Temsilcisi Global Desing Mimarlık Gaziantep +6 il daha	 Toprak Analizcisi Dajan Mardin	 Satış ve Pazarlama Süpervizörü Bat İstanbul(Avr.)
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- Search capability by Position and City. Both textboxes will support autocomplete. City search will show current city from browser location (or user citty (assume it is accessible))
- Show a list of at least 5 job postings on home page that are in the current city from browser location. Show other postings if there are none available
- 'Son Aramalarım' will show the recent searches made by the user
- If not logged in, login will be visible at the header.

SEARCH RESULTS

5 İzmir Yazılım Uzmanı İş İlanları

Önerilen

Ülke / Şehir / İlçe

Türkiye

1 şehir seçildi

İlçe seçin

Çalışma Tercihi

☐ İş Yerinde

☐ Uzaktan / Remote

☐ Hibrit

Uygula

Seçili Filtreler (3) Filtreleri Temizle

Yazılım Uzmanı X İzmir X Türkiye X

Yazılım Uzmanı

ALFEMO Alfemo

İzmir • İş Yerinde

Tam zamanlı

Bugün

Canias Yazılım Uzmanı

Packy Packy Packaging

İzmir • İş Yerinde

Tam zamanlı

Son gün

- Should be filtrable by a pane on the left at least by country, city, town and working preference
- Filters should be stated at the top of the list and can be deleted by an X if desired

Seçili Filtreler (3) Filtreleri Temizle

Yazılım Uzmanı X İzmir X Türkiye X

JOB POSTING DETAIL

Yazılım Uzmanı

Alfemo

Başvur

Kaydet



İzmir(Torbalı) • İş Yerinde

Bugün güncellendi

Çalışma Şekli

Tam Zamanlı

Pozisyon Seviyesi

Uzman

Departman

Bilgi Teknolojileri / IT

Başvuru Sayısı

282 başvuru

Yazılım Uzmanı

Alfemo

- Üniversitelerin ilgili bölümlerinden mezun (Bilgisayar Mühendisliği, Yazılım Mühendisliği veya Bilgisayar Programcılığı),
- Html5 , Css3, Bootstrap, Javascript, JQuery, Ajax, Json, Xml bilgisine sahip,
- Microsoft. NET platformu, C#, ASP.NET MVC ile uygulama geliştirme konusunda bilgi ve tecrübe sahibi,
- Tercihen MS SQL Server konusunda bilgi sahibi,
- Birimler arası entegre iş süreçleri hakkında bilgisi olan,

Daha Fazla Gör

Aday Kriterleri

Tecrübe: En az 2 yıl tecrübeli

Eğitim Seviyesi: Üniversite(Mezun), Yüksek Lisans(Öğrenci), Yüksek Lisans(Mezun)

Askerlik Durumu: Yapıldı, Muaf

İlgini Çekebilecek İlanlar



Backend Developer - Node.Js

VIGO TEKNOLOJİ VE LOJİSTİK A...

İstanbul(Asya) • Hibrit

Tam zamanlı

Bugün



Yazılım Uzmanı

DATA SİSTEM İŞE ALIM HİZMETL...

İstanbul(Avr.) • Hibrit

Tam zamanlı

Bugün

Search results will show

- Job posting detail will at least have description and location of posting, last updated date and number of applications
- On the right page, you will show at least 3 related postings that might interest the user
- Application (Başvur) button will be available. If user not logged in, user will be routed to login page.

NOTIFICATION SERVICE

User can create Job Posting Alerts (İs Alarmı) to be notified when new jobs are posted.

İş Alarmı Oluştur



Anahtar Kelimeler (1 Seçim)

Web Tasarım Uzmanı

Ülke / Şehir / İlçe

Türkiye

Şehir seçin

İlçe seçin

- write a job alert notification scheduled task
 - o to go over new job postings from a queue
 - o go over all user job posting alerts
 - o send to users if there are eligible postings.

- Write a related job notification scheduled task
 - o To go over job searches a user makes from a job searches a user made and send user notification about related job postings. Application will store job searches of user in a separate No SQL DB

AI AGENT SERVICE

Implement AI agent chat window in main application screen that will use APIs you created to perform search and apply use cases

 You: I'd like to search web development jobs in Istanbul.

 AI: Got it! Here are a few options you might like 🖱️

Junior web developer – Istanbul – Company 1

- Needs to be computer engineering graduate
- Need React and Angular skills
- [Apply]

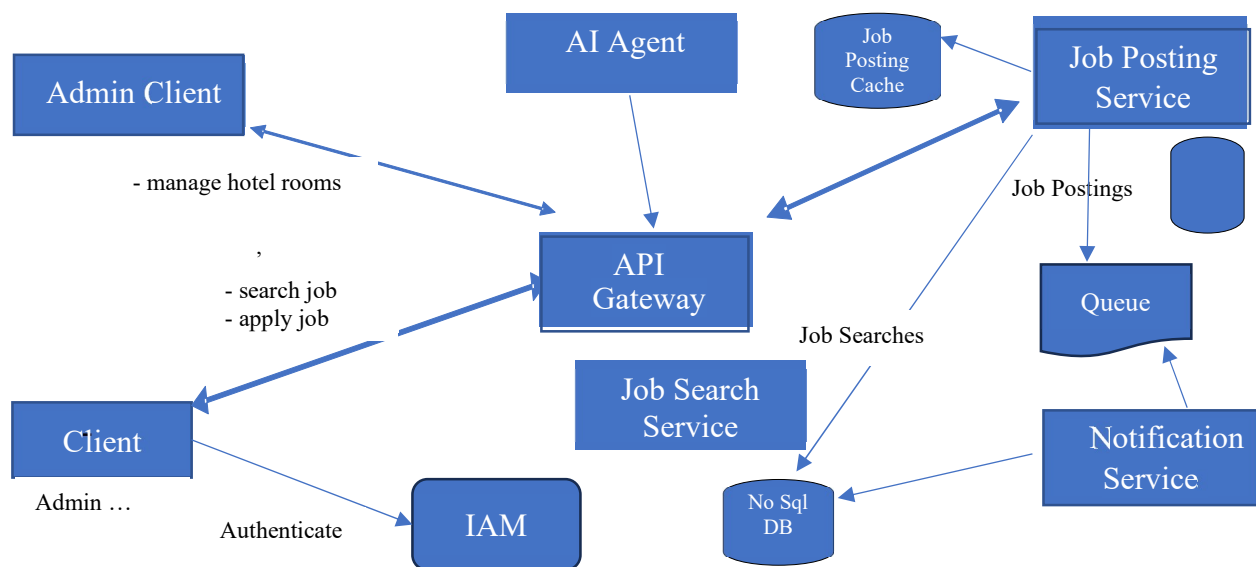
Frontend developer – Istanbul – Company 2

- Experienced in typescript
- Experience with Javascript components libraries
- [Apply]

 AI: Would you like to apply to any of these jobs?

☒ Yes, open details for **Frontend developer – Istanbul – Company 2**

NON-FUNCTIONAL REQUIREMENTS



- Job searches will be stored in a separate No SQL DB
- Job Postings will be stored in a separate distributed cache like Redis

Students

SILA BARIŞIK İSMET METE BAYKASOĞLU AYŞENUR TOMBUL ÖZGÜR CAN GÜNGÖR BERKER VERGİ SENA ASLI TÜRK EMRE AKAR	BEYZA TÜRK ÖMER FARUK AKKUŞ DİLARA KUYAR BAŞAK NUR KORKUT CANSU GÜN İPEK DEMİREL MELİS BABACAN YAGIZ YUNGUL
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COMMON REQUIREMENTS

- Project can be built on any service-oriented framework as long as requirements are met.
- Simple UI implementation per mock-ups given above is required. Front end UIs do not have to be same as shown above. It just needs to work
- All business use cases above must be available via REST webservice
- The project will be deployed to providers like Azure, Google Cloud, Vercel and AWS or any other cloud vendor
- Per deployment diagrams above, Job Search, Hotel and Notification services will be deployed separately. All APIs will be reached via an API gateway
- You can use RabbitMQ or Azure Messaging for queue solutions.
- All REST services must be versionable and support pagination when needed
- You can choose any development environment you like as long as they support REST services and is deployable in cloud.
- For caching, you can use services like Redis or use in-memory-caching
- All user authentication will be stored in an IAM service like AWS Cognito, Firebase Authentication or Supabase Auth. No external authentication like Google needed. Your local authentication implementation will not be accepted
- For AI Agent, real time messaging IS NOT required
- You can make assumptions as long as you document them
- At least one distributed caching solution needs to be implemented i.e Hotel Details
- MAKE SURE TO CREATE A DOCKER FILE in your source. See an example at <https://www.digitalocean.com/community/tutorials/how-to-build-a-node-js-application-with-docker-for-a-node.js> application. DO NOT CREATE A DOCKER IMAGE FILE, it would be too big and not needed
- DEPLOYMENT
 - o create a data model for each data source and use a DATABASE service from any cloud service you like i.e Azure SQL Server, Azure Cosmos Table. SQLite is NOT allowed
 - o For API and UI layer hosting of app services, use a cloud service i.e Azure App Services

- For API gateway, you can use the gateway you implemented in assignment. Azure API management is costly so refrain from using it unless you want to pay for it
- For queue service, use Azure Message Queues, Rabbit MQ
- For scheduling services, use a cloud service i.e Azure App Logic or <https://cloud.google.com/scheduler/>

DELIVERABLES

- Public Github repository link
- A README document in your GITHUB repository that has
 - Final deployed urls of the application
 - your design, assumptions, and issues you encountered.
 - Data models (i.e an ER)
 - Include a link to a short video (max 5 minutes) presenting your project.